

2022 ET OHIKOA

A4 $\int \ln(x^2-1) \cdot dx = I$

2ATIKAKO FUNTZIOA

$$\int u dv = u \cdot v - \int v \cdot du$$

$$\begin{cases} u = \ln(x^2-1) \rightarrow du = \frac{2x}{x^2-1} dx \\ dv = dx \rightarrow v = dx \end{cases}$$

$$I = \ln(x^2-1) \cdot x - \int \frac{x \cdot 2x}{x^2-1} dx = x \cdot \ln(x^2-1) - 2 \int \frac{x^2}{x^2-1} dx$$

$$= x \cdot \ln(x^2-1) - 2 \int \left(1 + \frac{1}{x^2-1} \right) dx = \frac{x^2}{-x^2+1} \frac{(x^2-1)}{1}$$

$$= x \cdot \ln(x^2-1) - 2 \int dx - 2 \int \frac{1}{x^2-1} dx =$$

$$= x \cdot \ln(x^2-1) - 2x - \underbrace{2 \int \frac{1}{x^2-1} dx}_{I_2}$$

$$I_2 = \int \frac{1}{x^2-1} dx \parallel \frac{1}{x^2-1} = \frac{A}{x+1} + \frac{B}{x-1}$$

$$1 = A(x-1) + B(x+1)$$

$$x=1 \rightarrow 1 = 2B \rightarrow \boxed{B=1/2}$$

$$x=-1 \rightarrow 1 = -2A \rightarrow \boxed{A=-1/2}$$

$$I_2 = \int \frac{1}{x^2-1} dx = \int \frac{-1/2}{x+1} dx + \int \frac{1/2}{x-1} dx =$$

$$= \boxed{-\frac{1}{2} \ln|x+1| + \frac{1}{2} \ln|x-1|}$$

$$I = x \cdot \ln(x^2-1) - 2x + \underbrace{\ln|x+1| - \ln|x-1|}_{\ln\left|\frac{x+1}{x-1}\right|} + K$$

Besta modu batara

$$\begin{aligned}\int \ln(x^2-1) dx &= \int \ln[(x+1)(x-1)] dx = \\ &= \underbrace{\int \ln(x+1) dx}_{I_1} + \underbrace{\int \ln(x-1) dx}_{I_2}\end{aligned}$$

$$I_1) \begin{cases} u = \ln(x+1) \rightarrow du = \frac{1}{x+1} \\ dv = dx \rightarrow v = x \end{cases}$$

$\begin{array}{r} x \\ -x-1 \\ \hline -1 \end{array} \quad \begin{array}{r} \frac{x+1}{1} \\ \hline \end{array}$

$$I_1 = \ln(x+1) \cdot x - \int x \cdot \frac{1}{x+1} dx$$

$$= \ln(x+1) \cdot x - \int \left(1 - \frac{1}{x+1}\right) dx =$$

$$= \boxed{\ln(x+1) \cdot x - x + \ln(x+1)} + k$$

$$I_2) I_2 = \ln(x-1) \cdot x - \int x \cdot \frac{1}{x-1} dx$$

$\begin{array}{r} x \\ +1 \\ \hline 1 \end{array} \quad \begin{array}{r} \frac{x-1}{1} \\ \hline \end{array}$

$$= \ln(x-1) \cdot x - \int \left(1 + \frac{1}{x-1}\right) dx$$

$$I_2 = \boxed{\ln(x-1) \cdot x - x - \ln(x-1)} + k$$

$$I = \underbrace{\ln(x+1) \cdot x - x + \ln(x+1)}_{\sim} + \underbrace{\ln(x-1) \cdot x - x - \ln(x-1)}_{\sim} + k$$

$$= \boxed{\begin{array}{l} x \left(\underbrace{\ln(x+1) \cdot \ln(x-1)}_{\sim} \right) - 2x + \underbrace{\ln(x+1) - \ln(x-1)}_{\sim} + k \\ x \cdot \ln(x^2-1) \quad \ln \left| \frac{x+1}{x-1} \right| \end{array}}$$